Sensor Stress Detection

Implementation Plan

1. Project Goal

Build a stress detection system using physiological sensor data from the Empatica E4 wristband, mapped to physical hardware sensors: MAX30102 (HR), DS18B20 (TEMP), GSR (EDA), MPU6500 (ACC).

2. Dataset

- 28 subjects with E4 wristband data (debate recordings, ~10-15 min each)
- 4 sensors used: EDA (4Hz), HR (1Hz), ACC-xyz (32Hz), TEMP (4Hz)
- Annotations every 5 seconds: arousal, valence, nervous, angry (1-5 scale)
- Annotation source: Aggregated external annotations (consensus of 5 raters)

3. Stress Label: Composite Score (Option A)

```
score = arousal*0.35 + (6-valence)*0.35 + nervous*0.20 + angry*0.10
normalized = (score - 1) / 4     [range 0 to 1]
stress = 1 if normalized > 0.5, else 0
```

4. Physical Sensor Mapping

Hardware Sensor	Dataset Signal	Sampling Rate
GSR	EDA	4 Hz
MAX30102	HR	1 Hz
MPU6500	ACC (x,y,z)	32 Hz
DS18B20	TEMP	4 Hz

5. Model: XGBoost Baseline with LOSO CV

1. Load sensor data + annotations for 28 subjects
2. Align to 5-second windows using startTime from subjects.csv
3. Extract 34 handcrafted features per window:
   EDA: mean, std, min, max, range, slope, peaks, AUC, kurtosis, skew (10)
   HR: mean, std, min, max, range, delta (6)
   ACC: per-axis mean/std, magnitude stats, jerk, SMA (12)
   TEMP: mean, std, min, max, slope, range (6)
4. Train XGBoost with Leave-One-Subject-Out cross-validation
5. Evaluate: accuracy, F1, precision, recall, ROC-AUC

6. Feature Details (34 Total)

Feature	Description	Stress Relevance
eda_mean	Avg skin conductance	High = high arousal
eda_std	Variability	High = fluctuating arousal
eda_slope	Linear trend	Rising = increasing stress

eda_num_peaks	SCR count	More peaks = stress responses
eda_auc	Area under curve	Total activation level
hr_mean	Average heart rate	High HR = stress/arousal
hr_std	HR variability	Low HRV = stress
hr_delta	HR change in window	Rising = acute stress
acc_mag_mean	Motion magnitude	Fidgeting = nervousness
acc_jerk_mean	Motion abruptness	Sudden moves = anxiety
temp_mean	Skin temperature	Cooling = vasoconstriction
temp_slope	Temp trend	Dropping = stress onset

7. File Structure

```
gemini model/
  implementation_plan.md
  implementation_plan.pdf
  first model(baseline model)/
    config.py              - Hyperparameters & paths
    data_loader.py         - Load E4 CSVs + annotations
    feature_engineering.py - 34 handcrafted features
    stress_labels.py       - Composite stress score
    train.py               - XGBoost + LOSO CV pipeline
    evaluate.py            - Metrics, plots, results
    outputs/               - Saved results & plots
```

8. Future Phases

- Phase 2: Deep Learning (CNN + BiLSTM)
- Phase 3: Multimodal (sensor + facial video fusion)